ALEXANDER COSTER

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SUMMARY

Mechanical engineer and M.S. candidate experienced in vehicle dynamics, robotics integration, CNC automation, and additive manufacturing. Skilled in SolidWorks CAD, finite element analysis (ANSYS), MATLAB/Simulink/Simscape, and Python, with strengths in design-for-manufacturing (DFM, GD&T), electromechanical integration, and data analysis — taking systems from concept through prototyping, testing, and validation.

EDUCATION

University of Oklahoma — M.S., Mechanical Engineering — Expected May 2027 | GPA 4.0

University of Oklahoma — B.S., Mechanical Engineering — May 2026 | GPA 3.41

EXPERIENCE

Sooner Advanced Manufacturing Lab — *Research Engineer* — Mar 2025 – Aug 2025

- Designed and built a robotic lathe automation cell: integrated an xArm6 robotic arm and pneumatics with a Tormach 8L CNC lathe for fully automated machining of titanium (Ti-6Al-4V) additive samples.
- Developed motion sequencing, pneumatic actuation, and safe robot-to-machine handoff logic, reducing manual machine-tending and improving part-to-part repeatability.

Undergraduate Research Assistant — AME Machine Shop, Univ. of Oklahoma — Jun 2023 – Mar 2025, Sep 2025 – Present

- Designed and built research apparatus and tooling for faculty and student projects, consulting on design-for-manufacturing.
- Manufactured components on CNC mills and manual equipment, holding tolerances using GD&T and CAM (HSMWorks).
- Provided SolidWorks CAD design support across multiple research groups.

FSAE Sooner Racing Team — Mechanical Engineer — Aug 2022 – Present

- Design and manufacturing of the frame, suspension, brakes, and ergonomics for the Formula SAE race car.
- Built a full-vehicle SolidWorks CAD model and a Simscape Multibody vehicle-dynamics model in MATLAB to analyze suspension kinematics and validate the vehicle design.

FSAE Sooner Racing Team — *Chassis Lead* — May 2023 – May 2024

- Led the chassis subsystem (frame, suspension, brakes, ergonomics, and safety) through design and manufacturing.

PROJECTS

Unitree G1 Humanoid Robot — robotics rebuild & AI integration — Summer 2026

- Rebuilt and recalibrated a Unitree G1 humanoid (per-joint calibration, actuator tuning, restored balance and locomotion) and integrated the OpenAI API for real-time conversational interaction.

VN-300 IMU Data & Analysis Tools — vehicle-dynamics software — 2026 – Present

- Building a Python data-acquisition and analysis suite for the VectorNav VN-300 GNSS/INS to log and process vehicle-dynamics data (speed, lateral/longitudinal acceleration, GPS timing) that informs suspension and handling design.

Ceramic Powder-Bed Fusion Printer — additive manufacturing R&D — 2024 – 2025

- Developed a custom powder-bed fusion printer: mechanical design, motion control (Klipper firmware), powder delivery, and laser integration for selective sintering.

SKILLS

CAD & Design:	SolidWorks, GD&T, DFM, tolerance analysis, tube-frame design, prototyping
Analysis & Simulation:	ANSYS (FEA), MATLAB, Simulink, Simscape Multibody, vehicle dynamics
Manufacturing:	CNC machining, CAM (HSMWorks), additive manufacturing, TIG/MIG welding, fixture & tooling design
Robotics & Controls:	robotic integration (xArm6, Unitree G1), ROS2/MoveIt, motion control, pneumatics, IMU/GNSS sensor integration
Programming & Tools:	Python, C++, Java, OpenAI API, Git/GitHub, GitLab, Linux